Blender

Model, Animate & Render in 3D

The free and open-source 3D creation suite

Interface, modeling, materials, lighting, rendering, animation, export, and a full mini project

Model | Animate | Render

OVERVIEW

What Is Blender?

Blender is a **free, open-source 3D creation suite** maintained by the Blender Foundation. In one program you can model objects, sculpt, texture, rig characters, animate, simulate physics, render photorealistic images, edit video, and even make 2D animation. It is used by hobbyists, indie game developers, architects, and professional studios. Download it from **blender.org** (Windows, macOS, and Linux). The interface and some menu names change slightly between versions, but the core tools and shortcuts in this guide are the same.

What can you make?

Area	Examples
3D modeling	Characters, props, furniture, buildings, vehicles
Sculpting	Organic shapes: creatures, faces, terrain
Animation	Character motion, product animations, logos, motion graphics
Rendering	Photorealistic stills, product shots, architectural visuals
Game and web assets	Models exported for Unity, Unreal, Godot, Three.js, and Babylon.js
3D printing	Models exported as STL files
Simulation and VFX	Smoke, fire, water, cloth, particles, camera tracking

The 3D workflow (pipeline)

Stage	What happens	Typical tools
1. Model	Build the shape	Edit Mode, modifiers, sculpting
2. UV unwrap	Flatten the surface so textures fit	UV Editor
3. Material	Add color, metal, glass, texture	Shader Editor, Principled BSDF
4. Rig and animate	Add bones and movement	Armatures, keyframes, Timeline
5. Light and camera	Set up the scene	Lights, HDRI, camera
6. Render / export	Produce final image, video, or game asset	EEVEE, Cycles, glTF export

Hardware tip: Blender runs on most modern computers, but a dedicated graphics card (GPU) and 16 GB of RAM make rendering and sculpting much smoother. Use a three-button mouse if you can; it makes navigation far easier.

1 Interface & Navigation

Find your way around the 3D viewport

Main areas of the screen

Area	What it does
3D Viewport	The big central window where you view and edit your scene
Outliner (top right)	List of every object, light, and camera in the scene; like a file explorer
Properties (bottom right)	Settings for the selected object, modifiers, materials, render, and world
Timeline (bottom)	Animation frames, playback, and keyframes
Toolbar (T, left)	Select, move, rotate, scale, and other tools
Sidebar (N, right of viewport)	Exact location, rotation, and scale values; view settings
Workspace tabs (top)	Layout, Modeling, Sculpting, UV Editing, Texture Paint, Shading, Animation, Rendering, Compositing, Geometry Nodes, Scripting

Navigating the 3D view

Action	Mouse / key
Orbit (rotate the view)	Hold middle mouse button and drag
Pan (slide the view)	Shift + middle mouse drag
Zoom	Scroll wheel (or Ctrl + middle mouse drag)
Front / Right / Top view	Numpad 1 / 3 / 7 (Ctrl + number for the opposite side)
Camera view	Numpad 0
Perspective / Orthographic	Numpad 5
Focus on the selected object	Numpad . (period)
Show everything	Home
No numpad? Use the pie menu	Press the ` (backtick) key, or enable Emulate Numpad in Preferences

Selecting and transforming objects

Action	Shortcut
Select / add to selection	Left click / Shift + left click
Select all / none	A / Alt + A
Box select	B (or drag with the Select Box tool)
Move (Grab)	G
Rotate	R
Scale	S
Lock to an axis	Press X, Y, or Z after G, R, or S (e.g., G then Z moves only up/down)
Type exact values	G, X, then type 2, then Enter moves 2 units along X
Confirm / Cancel	Enter or left click / Esc or right click

> **Always remember:** `Ctrl+Z` undoes mistakes and `Ctrl+S` saves your **.blend** file. Save often, because Blender does not save automatically unless you enable auto-save in Preferences.

Adding and managing objects

Action	Shortcut
Add a new object (mesh, light, camera...)	Shift + A
Delete	X or Delete
Duplicate	Shift + D (then move or press Esc)
Search any command	F3
Switch Object Mode / Edit Mode	Tab
Hide / reveal	H / Alt + H
Rename an object	F2
Parent selected to active	Ctrl + P
Join objects together	Ctrl + J

2 Modeling

Build shapes from vertices, edges, and faces

Mesh basics

Most 3D models are **meshes** made of **vertices** (points), **edges** (lines between points), and **faces** (surfaces). Press **Tab** to enter **Edit Mode** and change the shape. Press **1**, **2**, or **3** to switch between vertex, edge, and face select modes.

Essential Edit Mode tools

Tool	Shortcut	What it does
Extrude	E	Pull out new geometry from selected faces, edges, or vertices
Inset faces	I	Create a smaller face inside the selected face
Bevel	Ctrl + B	Round or chamfer sharp edges
Loop cut	Ctrl + R	Add a ring of edges around the model
Knife	K	Cut new edges freehand
Merge	M	Combine vertices into one point
Fill	F	Create a face from selected vertices or edges
Select edge loop	Alt + click	Select a whole ring of edges
Separate part	P	Split selection into its own object

Modifiers: powerful non-destructive tools

Modifiers change an object without permanently altering the mesh. Add them in **Properties > Modifiers (wrench icon) > Add Modifier**.

Modifier	Use
Subdivision Surface	Smooths and rounds a low-poly shape (Ctrl + 1/2/3 sets the level quickly)
Mirror	Model one half; the other half is mirrored automatically
Array	Repeat an object many times (fences, stairs, chains)
Bevel	Adds rounded edges that catch light realistically
Solidify	Gives thickness to a flat surface (cloth, walls, cups)
Boolean	Cut one shape from another, or combine shapes
Decimate	Reduces polygon count for game and web models

Modeling best practices

- Start simple with a primitive (cube, cylinder, sphere) and refine it step by step
- Keep faces as **quads** (four-sided) for clean smoothing and animation
- Apply **Scale** (Ctrl + A > Scale) before using modifiers so they behave correctly
- Use **Object > Shade Smooth** for rounded surfaces
- Turn on **Statistics** in the viewport overlays to watch your polygon count

3 Materials, UVs & Lighting

Make your models look real

Materials with the Principled BSDF

Select an object, open **Properties > Material**, and click **New**. The Principled BSDF shader controls most real-world surfaces:

Setting	Meaning	Example
Base Color	The main color or image texture	Red paint
Metallic	0 = non-metal, 1 = metal	1.0 for gold or steel
Roughness	0 = mirror shiny, 1 = fully matte	0.1 for glossy plastic, 0.8 for wood
Transmission	Makes the surface transparent like glass	1.0 for a glass bottle
Emission	Makes the surface glow	A lamp or neon sign

Adding image textures

- Open the **Shading** workspace; the Shader Editor appears at the bottom
- Press **Shift + A > Texture > Image Texture** and load an image
- Connect its **Color** output to the Principled BSDF **Base Color** input
- Free realistic textures and HDRIs: **Poly Haven**, **ambientCG**

UV unwrapping (so textures fit)

In Edit Mode, press **A** to select all, press **U**, and choose **Smart UV Project** (quick) or **Unwrap** (after marking seams with Ctrl + E > Mark Seam). View the result in the **UV Editing** workspace.

Lights and the world

Light type	Best for
Point	Light bulbs, candles; shines in all directions
Sun	Outdoor daylight; parallel rays and sharp shadows
Spot	Stage lights, flashlights; cone of light
Area	Soft studio lighting; the most useful for product renders

Three-point lighting: a bright **key light** from the front side, a softer **fill light** from the other side, and a **rim light** behind the subject to outline it. For natural lighting, use **Properties > World > Color > Environment Texture** and load an HDRI image.

4 Camera & Rendering

Produce your final images and videos

Setting up the camera

- Add a camera with **Shift + A > Camera**, then press **Numpad 0** to look through it
- Position it by enabling **View > Align View > Align Active Camera to View** (Ctrl + Alt + Numpad 0) after framing a shot in the viewport
- Adjust **Focal Length** in the camera properties (35 mm is natural, 85 mm for portraits)
- Enable **Depth of Field** for a blurred background

EEVEE vs. Cycles

	EEVEE	Cycles
Type	Real-time rasterizer	Path tracer (simulates real light)
Speed	Very fast; seconds per image	Slower; minutes per image
Realism	Good, great for stylized work	Excellent, photorealistic
Best for	Previews, animation, games, motion graphics	Final stills, glass, accurate reflections
Hardware	Needs a decent GPU	Runs on CPU or GPU (set in Preferences)

Output settings and rendering

Step	Where / shortcut
Choose the render engine	Properties > Render > Render Engine
Set resolution (e.g., 1920 x 1080)	Properties > Output > Resolution
Cycles quality: samples and Denoise	Properties > Render > Sampling (enable Denoise)
Render a still image	F12, then Image > Save As to export PNG or JPEG
Render an animation	Set Frame Start / End, choose FFmpeg Video (MP4, H.264) in Output, then Ctrl + F12
Look in the viewport while working	Viewport shading: Solid, Material Preview, or Rendered (top right of the viewport)

> **Tip:** Render animations as an **image sequence (PNG)** first. If Blender crashes, you can resume instead of losing everything; then combine frames into a video.

5 Animation & Sculpting

Bring objects to life and shape organic forms

Keyframe animation in 5 steps

Step	Action
1	Go to the **Animation** workspace and set the frame rate in Output properties (24 or 30 fps is common)
2	On frame 1, select the object and press **I** to insert a keyframe (Location, Rotation, or Scale)
3	Move the playhead in the Timeline to frame 60
4	Move or rotate the object, then press **I** again
5	Press **Space** to play the animation; Blender fills in the motion between keyframes

Animation editors

Editor	Purpose
Timeline	Quick playback and keyframe overview
Dope Sheet	Move, copy, and time keyframes for many objects
Graph Editor	Fine-tune motion curves for smooth, natural easing
Armatures (bones)	Rig characters so you can pose and animate limbs
Shape Keys	Morph a mesh, e.g., facial expressions

Sculpting basics

Switch to the **Sculpting** workspace to shape a mesh like clay. Useful brushes:

Brush	Use
Draw / Clay Strips	Build up volume
Grab	Pull and move large areas
Smooth (hold Shift)	Soften bumps
Inflate / Crease	Add puffiness or sharp lines

Adjust brush size with **F** and strength with **Shift + F**. Use **Remesh** (Ctrl + R in Sculpt Mode) or the **Multiresolution** modifier to get enough detail to sculpt.

6 Python Scripting & Exporting

Automate Blender and send models to the web and games

Scripting with Python (bpy)

Blender has a built-in Python API. Open the **Scripting** workspace, paste code in the Text Editor, and click **Run**. This example creates a red cube:

```
import bpy

bpy.ops.mesh.primitive_cube_add(size=2, location=(0, 0, 1))
cube = bpy.context.active_object
cube.name = "MyCube"

mat = bpy.data.materials.new("Red")
mat.use_nodes = True
bsdf = mat.node_tree.nodes["Principled BSDF"]
bsdf.inputs["Base Color"].default_value = (1, 0, 0, 1)   # R, G, B, Alpha
cube.data.materials.append(mat)
```

Export formats

Format	Use it for
glTF 2.0 (.glb)	**Web 3D** (Three.js, Babylon.js), AR, and many game engines. Best choice for the web
FBX (.fbx)	Unity, Unreal Engine, animations and rigs
OBJ (.obj)	Simple static models, widely supported
STL (.stl)	3D printing
USD / Alembic	Film pipelines, large scenes, and cached animation

Preparing a model for the web (GLB workflow)

Step	What to do
1	Keep the polygon count low (use Decimate if needed) and use small texture sizes (1024 or 2048 px)
2	Apply all transforms: select the object and press Ctrl + A > All Transforms
3	Use Principled BSDF materials only, because glTF supports them well
4	Choose **File > Export > glTF 2.0 (.glb/.gltf)** and select **glTF Binary (.glb)**
5	Load it in Three.js with `GLTFLoader` or in Babylon.js with `SceneLoader`

PUTTING IT ALL TOGETHER

Mini Project: A Coffee Mug Scene

A classic beginner project that uses modeling, modifiers, materials, lighting, and rendering. Follow the steps in order.

#	Step	How
1	Start fresh	Open Blender, click on the default cube, press X to delete it
2	Add the cup body	Shift + A > Mesh > Cylinder. Press Tab, press 3 for face mode, select the top face, then press X > Faces to delete it (this opens the cup)
3	Give it thickness	Back in Object Mode, add a Solidify modifier (Properties > Modifiers) and set the thickness to about 0.05
4	Smooth it	Add a Subdivision Surface modifier (Ctrl + 2) and use Object > Shade Smooth. Add Loop Cuts (Ctrl + R) near the rim and base to keep edges crisp
5	Make the handle	Shift + A > Mesh > Torus. Scale it thin (S), rotate with R, and place it on the side; later join with Ctrl + J
6	Add a table	Shift + A > Mesh > Plane. Scale it up (S) and place it under the mug
7	Create materials	Ceramic: Base Color white or blue, Roughness 0.2. Table: wood texture, Roughness 0.6
8	Light the scene	Add an Area light above and in front; increase its size for soft shadows. Optionally add an HDRI in World properties
9	Set the camera	Shift + A > Camera, press Numpad 0, and frame a pleasing angle with Ctrl + Alt + Numpad 0
10	Render	Properties > Render > Cycles, enable Denoise, press F12, then Image > Save As

Level up the project

- Model a saucer and a spoon, and add steam with a particle system
- Animate a slow camera orbit and render an MP4, or export the mug as .glb for Three.js

TROUBLESHOOTING

Common Problems and Fixes

Problem	Likely cause	Fix
I cannot see my object	It is off-screen or hidden	Press Home or Numpad . ; check the Outliner for hidden objects (Alt + H)
Everything looks flat or black in Material Preview	No light or world lighting	Add a light or an HDRI in World properties
Pink or magenta textures	Image file is missing	File > External Data > Find Missing Files, or re-link the image
Modifier looks stretched or odd	Scale is not applied	Ctrl + A > Scale in Object Mode
Strange dark shading on faces	Flipped normals	Edit Mode > Mesh > Normals > Recalculate Outside (Shift + N)
Viewport is slow	Too many polygons or a high Subdivision level	Lower Viewport Levels; use Solid mode while modeling
Render is noisy	Too few samples	Raise samples and enable Denoise
Render is very slow	CPU rendering or high settings	Enable GPU in Preferences > System; lower resolution or samples while testing
Keyframes do not move anything	Wrong property keyed or playhead at the wrong frame	Check the Timeline and Graph Editor
Exported model has no textures	Textures not packed or unsupported nodes	Use Principled BSDF with image textures and export .glb

Safety tip: Use **File > Save Incremental** or save versions (mug_v1.blend, mug_v2.blend) so you can always go back, and keep backups of important projects.

NEXT STEPS

Learning Roadmap & Shortcut Cheat Sheet

Stage	Focus	Practice project
1. Interface	Navigation, G/R/S, selection, saving	Arrange primitives into a simple scene
2. Modeling	Edit Mode, extrude, loop cut, modifiers	Mug, chair, low-poly house
3. Materials	Principled BSDF, textures, UV unwrap	Textured crate or wooden table
4. Lighting and rendering	Lights, HDRI, camera, EEVEE and Cycles	Product shot of a bottle
5. Animation	Keyframes, Graph Editor, armatures	Bouncing ball, then a rigged character
6. Sculpting	Brushes, remesh, retopology	Stylized creature bust
7. Advanced	Geometry Nodes, simulations, Python, game and web export	Procedural scene or a web 3D viewer

Top shortcuts

Shortcut	Action	Shortcut	Action
G / R / S	Move / Rotate / Scale	Shift + A	Add object
Tab	Edit Mode toggle	Shift + D	Duplicate
E	Extrude	X	Delete
I	Inset / Keyframe	Ctrl + R	Loop cut
F12	Render image	Ctrl + F12	Render animation
Ctrl + Z	Undo	Ctrl + S	Save

Best practices and resources

- Practice a little every day and learn a few shortcuts per week
- Name your objects, organize them into collections, and use reference images
- Learn more: **docs.blender.org/manual** (official), **Blender Studio**, **Blender Guru** and **CG Cookie** tutorials on YouTube, and free assets on **Poly Haven**

> **Remember:** Every great 3D artist started with a single cube. Finish small projects, share them, and improve with each one.